Sri Chaitanya IIT Academy., India.

**A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI**

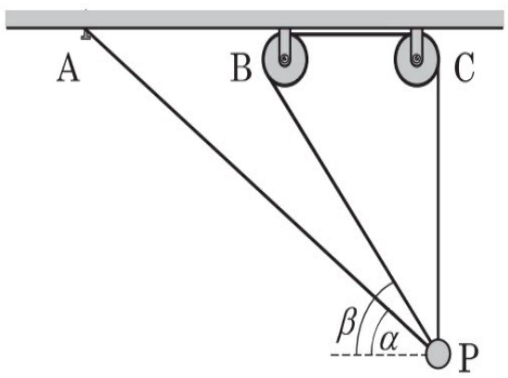
**A right Choice for the Real Aspirant**

**Regional Office: Chennai - Tamilnadu**

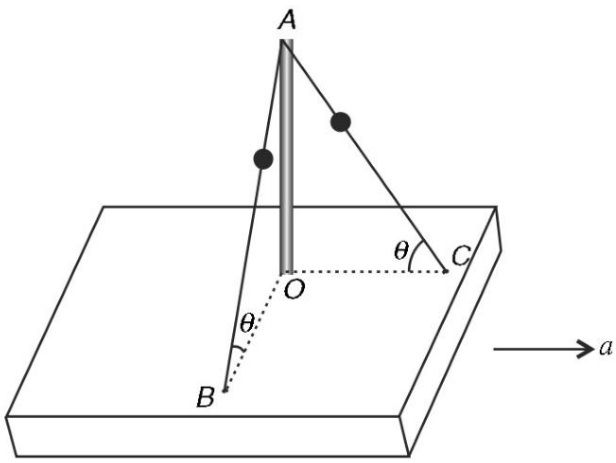
**MECHANICS-01** DAY-01 (15-03-24)

1. One end of alight inextensible cord is attached to a nail A on the ceiling. The cord passes through frictionless hole of a small bead P thereafter it passes rough two ideal pulleys and finally the other end is attached to the bead. Initially the bead is held at rest with the cord segments AP and BP making angles $\alpha$ and $\beta$ with the horizontal while keeping the segment CP vertical as shown in the figure. acceleration of the bead immediately after it is released is g/n then n is _____

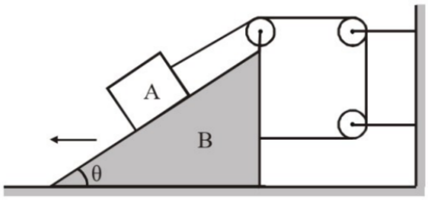
Acceleration due to gravity is g. (Take $\alpha = 30^0$ $\beta = 60^0$)




2. A horizontal wooden block has a fixed rod OA standing on it. From top point A of the rod, two wires have been fixed to points B and C on the block. The plane of triangle OAB is perpendicular to the plane of the triangle OAC. There are two identical beads on the two wires. One of the wires is perfectly smooth while the other is rough. The wooden block is moved with a horizontal acceleration (a) that is perpendicular to the line OB and it is observed that both the beads do not slide on the wire. The minimum coefficient of friction between the rough wire and the bead is $\dfrac{2n\sin\theta}{\sqrt{\cos^2\theta + \tan^2\theta}}$ then n is _______



3. A block A and wedge B are connected through a string as shown. The wedge B is moving away from the wall with acceleration $2m/s^2$ horizontally and acceleration of block A is vertically upwards as seen from ground. Then (no friction is present, pulleys and strings are mass less)



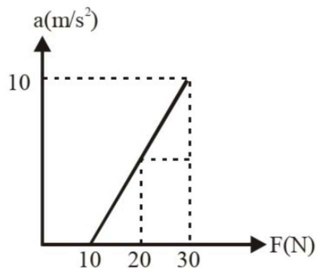

A) Acceleration of A with respect to B is $4m/s^2$

B) Acceleration of A with respect to B is $2\sqrt{3}m/s^2$

C) Angle $\theta$ is $60^0$

D) Acceleration of A is $2\sqrt{3}\,m/\sec^2$ w.r.t ground

4. A block placed on a rough horizontal surface is pushed with a force F acting horizontally on the block. The magnitude of F is increased and acceleration produced is plotted in the graph shown.



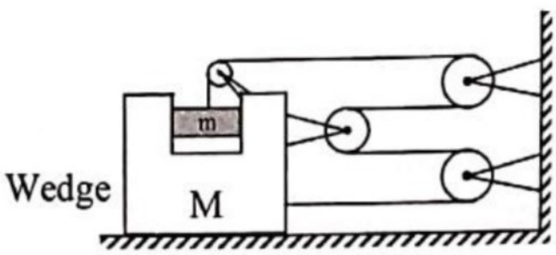

A) Mass of the block is 2 kg.

B) Coefficient of friction between block and surface is 0.5.

C) Limiting friction between block and surfaces is 10 N

D) When $F = 8N$, friction between block and surface is 10 N.

5. In the given arrangement string and pulleys are massless and mass of wedge M = 4m, (m – mass block) then which of the following statements are correct? (Neglect friction every where)



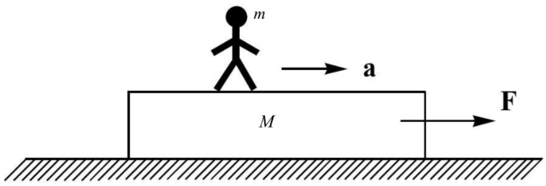
A) Downward acceleration of m with respect to wedge is $\frac{16g}{21}$

B) Acceleration of wedge with respect to ground is $\frac{4g}{21}$

C) Acceleration of mass m with respect to ground is $\frac{4g}{21}\sqrt{17}$

D) Acceleration of wedge with respect to ground is $\frac{5g}{21}$

6. A plank of mass M is placed on a rough horizontal surface. A man of mass m walks on the plank with an acceleration 'a' while the plank is also acted upon by a horizontal force F whose magnitude and direction can be adjusted to keep the plank at rest. Coefficient of friction between plank and surface is $\mu$. Choose the correct option(s)



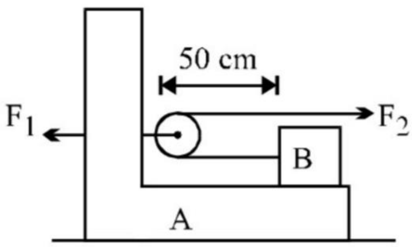

A) The maximum value of F to keep the plank at rest is $ma+\mu(M+m)g$

B) The minimum value of F to keep the plank at rest is $ma-\mu(M+m)g$

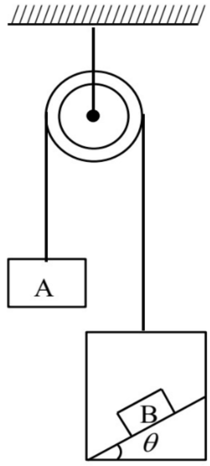
C) Direction of friction on the plank due to ground will always be towards right

D) If F = ma then friction on man due to plank is also F and friction between plank and ground is zero

7. A 1kg block B rests as shown on a bracket A of same mass. Constant force $F_1 = 20N$ and $F_2 = 8N$ start to act at time $t = 0$ when the distance of block B from pulley is 50cm. Time (in sec) when block B reaches the pulley is________(consider all surface are smooth)




8. In the arrangement shown, the string is light, pulley is smooth, the total mass on left hand side of the pulley is $m_1$, on right hand side is $m_2$, the coefficient of friction between block B and the wedge is $\mu = 0.8$ and the inclination angle $\theta = 30^0$. Select the **incorrect** option(s)



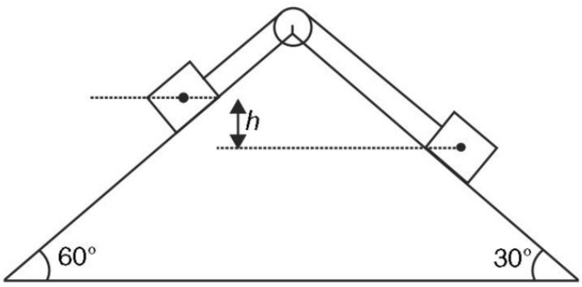

A) Block B will slide down if $m_1 = m_2$

B) Block B may remain stationary with respect to wedge only for suitable values of $m_1$ and $m_2$ with $m_2 > m_1$

C) Block B will slide down if $m_1 > m_2$

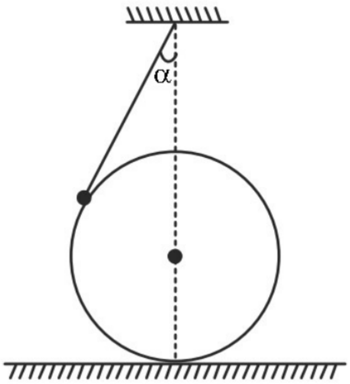
D) Block B can remain stationary with respect to wedge in any case

9. Two blocks of equal mass have been placed on two faces of a fixed wedge as shown in figure. The blocks are released from position where centre of one block is at a height $h = 20\,m$ above the centre of the other block. Find the time (in sec) after which the centre of the two blocks will be at same horizontal level. There is no firction anywhere.



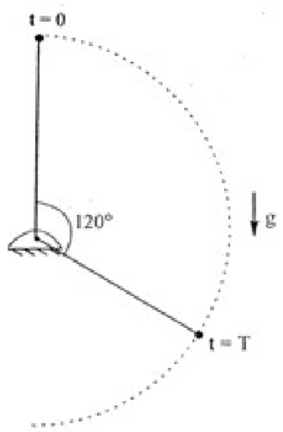

10. A sphere of mass M is held at rest on a horizontal floor. One end of a light string is fixed at a point that is vertically above the centre of the sphere. The other end of the string is connected to a small particle of mass $m$ that rests on the sphere. The string makes an angle $\alpha = 30^0$ with the vertical. Find the acceleration (in $m/\sec^2$) of the sphere immediately after it is released. There is no friction anywhere.

(Take $m = 2kg, M = 1kg, \sqrt{2} = 1.41\ \sqrt{3} = 1.73\ \sqrt{5} = 2.23, g = 10\,m/\sec^2$)




11. A light rod pendulum (of length $l$ with a small bob of mass m)is released from rest from its top position with a negligible push at t = 0. At t = T it swings through an angle of $120^o$ in the vertical plane. At this instant bob loses contact with the rod suddenly without changing its velocity. The magnitude of change in vertical component of acceleration of the bob just before t = T to just after t = T is $\frac{xg}{y}$. Then the value of $x-y$ is___

(where $x$ and y are smallest positive integers )




12. The power output from the motor of a 2 kg radio-controlled toy car is dependent on the velocity of the car and is given by $P = \frac{v(2-v)}{3}$ where P, v are in S.I units. Assume the car starts from rest and then the work done by all the forces in S.I units on the car by the time its motor power output is maximum will be

13. A hypothetical motor car ( mass m) is engineered and rated to be driven at a constant power $P_o$. It needs to start from rest and increase its speed against the action of a constant retarding force R. It was found to raise to a maximum speed $v_o$ in time T. Then

A) $v_o \to \infty$ B) $T \to \infty$ C) $v_o = \frac{P_o}{R}$

D) the time taken from start to rise to half the maximum speed is $\frac{mP_o}{R^2}\left[ln2 - \frac{1}{2}\right]$

14. The potential energy 'U' in Joule of a particle of mass 1kg, moving in X-Y plane, obeys the law $U = 3x + 4y$, where $(x, y)$ are the co-ordinates of the particle in meter. If the particle is at rest at $(6,4)$ at t=0, then (Assuming there are no non-conservative forces acting on the partcile)

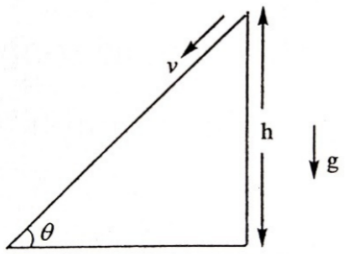
A) the particle has constant acceleration

B) the work done by the external forces, from the positon of rest of the particle to the instant of crossing the X-axis is 50J

C) the speed of the particle when it crosses the Y-axis is $10\,ms^{-1}$

D) the co-ordinates of the particle at time t=4s are $(-18,-28)$

15. In a mall, a travelator is motored to move down at a constant speed v for the convenience of people. Inclination and height associated are shown in figure. A person who can walk only at a constant speed $u$ relative to any surface decides to climb up this travelator.Assume height of the person is very small and he walks on the travelator without splipping his feet along the surface and reaches the top.Then




A) $u$ should be greater than v

B) Power delivered by friction to the person with respect to ground frame is zero.

C) Work done by internal forces of person on himself is $mgh\left(\frac{u}{u-\text{v}}\right)$

D) Work done by gravity on person with respect to ground frame is $-mgh$

16. A particle moves on acircular path of radius R, so that its angular velocity about a fixed point on the circumference of the circular path is constant and is equal to $\omega$. Then

A) the acceleration of the particle is variable.

B) the acceleration of the particle is constant in magnitude and is equal to $4R\omega^2$

C) the acceleration of the particle varies in magnitude only

D) acceleration of the particle will be zero.

**Question Stem for Question Nos.** 17 , 18

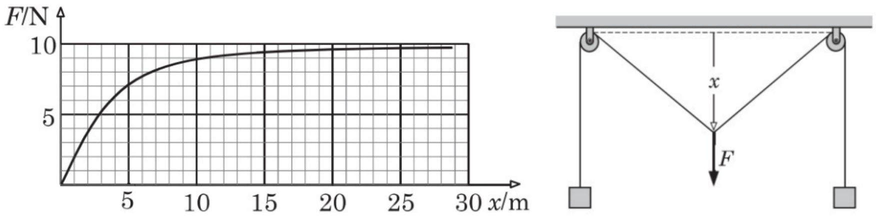
An engine can pull 4 coaches at a maximum speed of 20 m/s. Mass of the engine is twice the mass of every coach. Assuming resistive forces to be proportional to the weight, maximum speeds of the engine when it pulls 12 and 6 coaches are X and Y m/s respectively

17. Then value of X is________

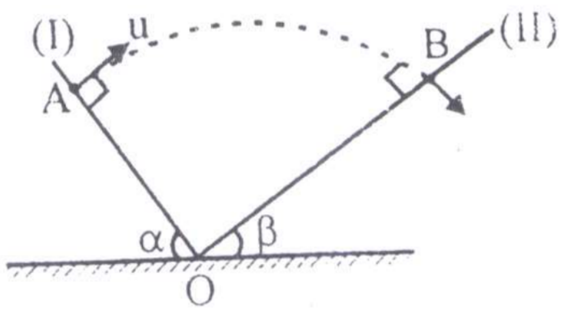
18. Then value of Y is________

19. A ball projected vertically upwards with a velocity $20\,m/s$ returns back on the ground with a velocity 16 m/s. If the air resistance is proportional to the speed of the ball, find airtime of the ball. Acceleration of free fall is $10\,m/s^2$.

20. A light inextensible cord supporting two identical loads at its ends passes over two ideal small pulleys fixed to a horizontal ceiling. If the midpoint of the cord between the pulleys is gradually pulled downwards, the pulling force F varies with displacement $x$ of the midpoint according to the following graph. Acceleration of free fall is $10\,m/s^2$. The distance between the pulleys is closest to




21. Two inclined planes **(I)** and **(II)** have inclination $\alpha$ and $\beta$ respectively with horizontal, (where $\alpha + \beta = 90^0$) intersect each other at point O as shown in figure. A particle is projected from point A with velocity u along a direction perpendicular to plane **(I).** If the particle strikes plane **(II)** perpendicular at B, then



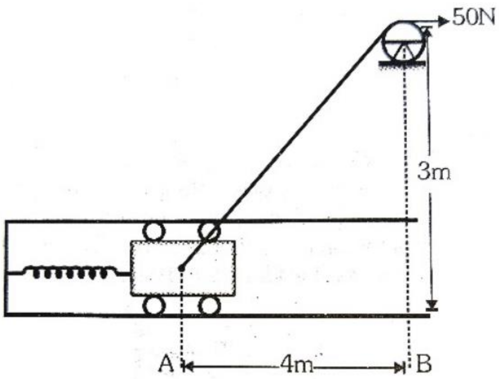

A) time of flight $= \dfrac{u}{g\sin\beta}$

B) time of flight $= \dfrac{u}{g\sin\alpha}$

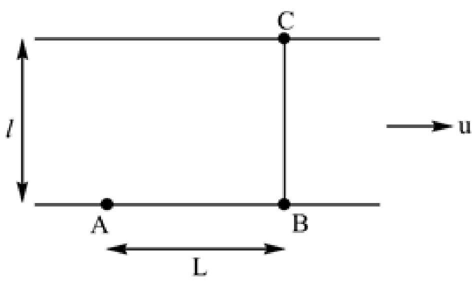
C) distance OB $= \dfrac{u^2}{2g\sin\beta}$

D) distance OB $= \dfrac{u^2}{2g\sin\alpha}$

22. A 20 kg block attached to a spring of force constant 5 N/m is released from rest at A. The spring at this instant is having an elongation of 1m. The block is allowed to move in smooth horizontal slot with the help of constant force of 50 N as shown. The velocity of block (in m/s) as it reaches B is ( Assume the rope to be light)




23. Two friends A and B are standing on a river bank L distance apart. They have decided to meet at a point C on the other bank exactly opposite to B. Both of them start rowing simultaneously on boats which can travel with velocity $V = 5\ km/hr$ in still water. It was found that both reached at C at the same time. Assume that path of both the boats are straight lines. Width of the river is $l = 3.0\,km$ and water is flowing at a uniform speed of $u = 3.0$ km/hr.




A) The time taken by two friends to cross the river is 1 hour

B) The time taken by two friends to cross the river is $\frac{3}{4}$ hour

C) $L = 6\ km$

D) $L = 4.5\ km$

24. A particle of mass m moves on the x-axis as follows : it starts from rest at t = 0 from the point x = 0, and comes to rest at t =1 at the point x = 1. No other information is available about its motion at intermediate times (0<t<1). If α denotes the instantaneous acceleration of the particle, then

A) α cannot remain positive for all t in the interval $0 \le t \le 1$

B) $|\alpha|$ cannot exceed 2 at any point in its path

C) $|\alpha|$ must be $\geq 4$ at some point or points in its path

D) $\alpha$ must change sign during the motion, but no other assertion can be made with the information given.

25. The speed of a train increases at a constant rate $\alpha$ from zero to $v$ and then remains constant for an interval and finally decreases to zero at a constant rate $\beta$. The total distance travelled by the train is $l$. The time taken to compete the journey is $t$. Then

A) $t=\frac{l(\alpha+\beta)}{\alpha\beta}$

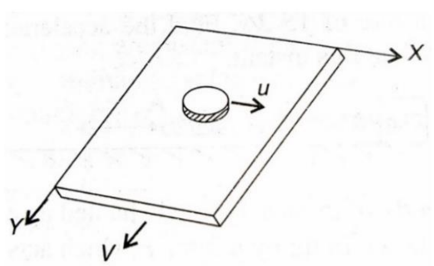
B) $t=\frac{l}{v}+\frac{v}{2}\left(\frac{1}{\alpha}+\frac{1}{\beta}\right)$

C) $t$ is minimum when $v=\sqrt{\frac{2l\alpha\beta}{(\alpha-\beta)}}$

D) $t$ is minimum when $v=\sqrt{\frac{2l\alpha\beta}{(\alpha+\beta)}}$

**Question Stem for Question Nos. 26 and 27:**

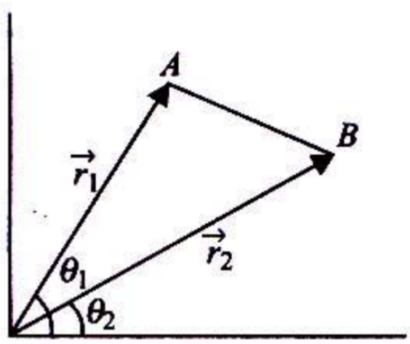
A large flat board is moving with $\vec{V}=2\hat{j}\ \text{ms}^{-1}$ on a smooth ground. A disc of mass $m=2kg$ moving with $\vec{u}=2\hat{i}\ ms^{-1}$ enters onto the board. The coefficient of friction between the disc and the board is $\mu=0.2$. If the disc and the board are to be moved with same constant velocities by external forces(velocities given in reference frame of the ground). The power of external force applied on the disc is X W and the power of external force applied on the board is Y W . (take $g=10\ \text{ms}^{-1}$ and $\sqrt{2}=1.41$)



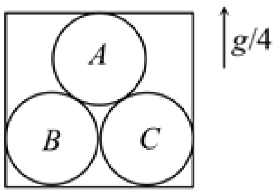

26. Then X value is _____

27. Then Y value is _____

28. In a two dimensional motion of a particle, the particle moves from point A, position vector $\vec{r}_1$, to point B, position vector $\vec{r}_2$. If the magnitudes of these vectors are respectively, $r_1=3\text{m}$ and $r_2=4\text{m}$ and the angles they make with the $x$–axis are $\theta_1=75^0$ and $\theta_2=15^0$, respectively, then find the magnitude of the displacement vector( in multiples) of $\sqrt{13}\text{m}$



29. Three identical spheres, each of mass $m$, are kept in contact inside a box as shown in figure. If box is moving vertically upward with an acceleration g/4, then (neglect friction)




(Neglect the force between sphere A and top surface of box)

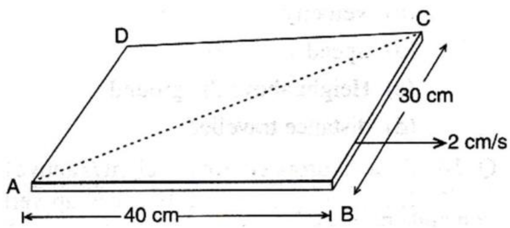
A) Normal force applied by the spheres on the bottom of the box is $\frac{9}{4}mg$

B) Normal force applied by the spheres on the bottom of the box is $\frac{15}{4}mg$.

C) Normal force between spheres A and B is $2\sqrt{3}mg$.

D) Normal force between spheres A and B is $\frac{5mg}{4\sqrt{3}}$.

30. A rectangular cardboard ABCD has dimensions of $40\,cm \times 30\,cm$. It is moving in a direction perpendicular to its shorter side at a constant speed of 2cm/s. A small insect starts at corner A and moves to diagonally opposite corner C (in a straight line with respect to cardboard along AC). On reaching C it immediately turns back and moves to A (in a straight line with respect to cardboard along CA). Throughout the motion the insect maintains a constant speed relative to the board. It takes 10s for the insect to reach C starting from A. The distance travelled by the insect in the reference frame attached to the ground, in the interval the insect starts from A and comes back to A is (in cm) $a\left(6\sqrt{5}+2\sqrt{13}\right)$. Then a = ___________




**KEY:**

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|---|---|---|---|---|---|
| 1-2 | 2-0.5 | 3-ACD | 4-ABC | 5-ABC | 6-ABD |
| 7-0.5 | 8-ABC | 9-4 | 10-3.46 | 11-3 | 12-1 |
| 13-BCD | 14-ACD | 15-ACD | 16-AB | 17-8.57 | 18-15 |
| 19-3.6 s | 20-10 m | 21-AC | 22-2 | 23-BD | 24-AC |
| 25-BD | 26-5.64 | 27-5.64 | 28-1 | 29-BD | 30-5 |